

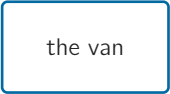
Reinforcement Learning

from one delivery van to the Bellman equation

Carlos Cotrini, ETH Zurich

Motivation

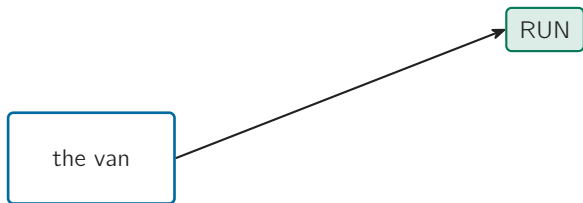
One van, one call a week



the van

every week: exactly one call

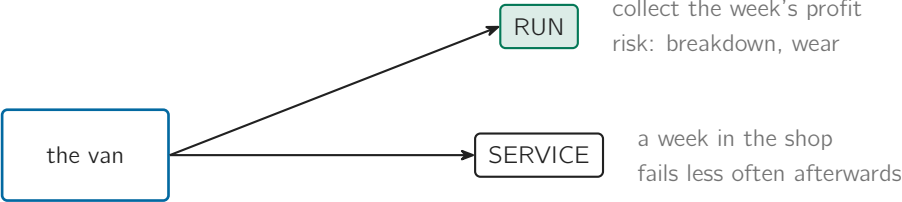
One van, one call a week



collect the week's profit
risk: breakdown, wear

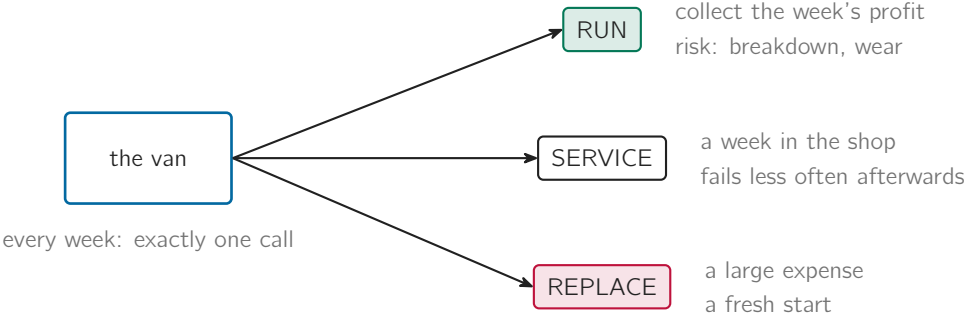
every week: exactly one call

One van, one call a week

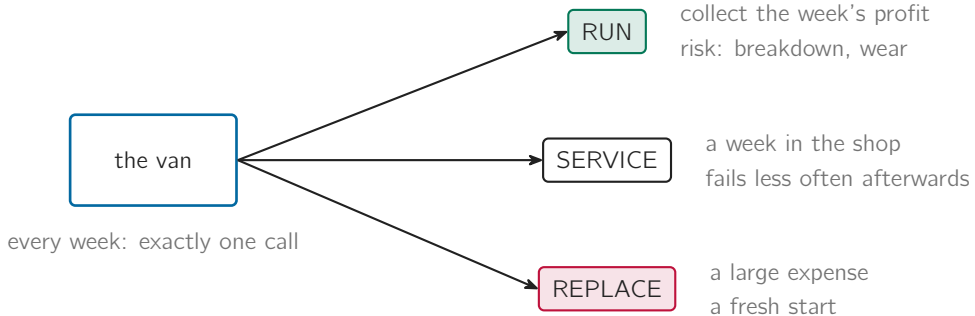


every week: exactly one call

One van, one call a week



One van, one call a week



short-term cost against long-term payoff, chance at every step

Markov decision processes

The model: four ingredients

S: states

four wear
levels

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S: states

four wear
levels

A: actions

three
calls

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T : dynamics

how the world
reacts

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$\gamma = 0.9$

how much the
future counts

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how much the
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$$(s', r) = T(s, a)$$

The model: four ingredients

S : states

four wear
levels

A : actions

three
calls

T : dynamics

how the world
reacts

$\gamma = 0.9$

how much the
future counts

$$\left(\underbrace{s'}_{\text{the state next week}}, \underbrace{r}_{\text{the reward}} \right) = T(s, a)$$

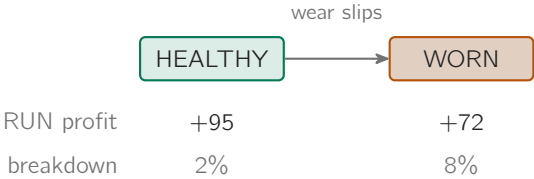
The fleet sheet

HEALTHY

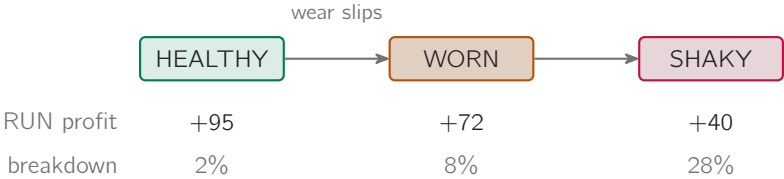
RUN profit +95

breakdown 2%

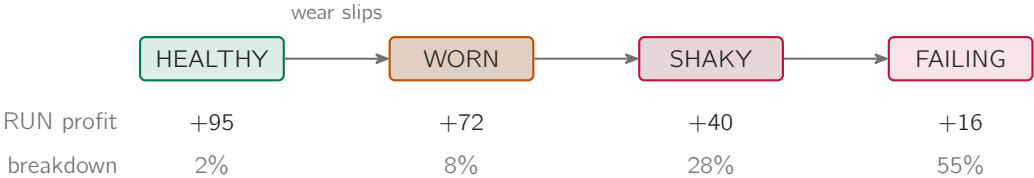
The fleet sheet



The fleet sheet



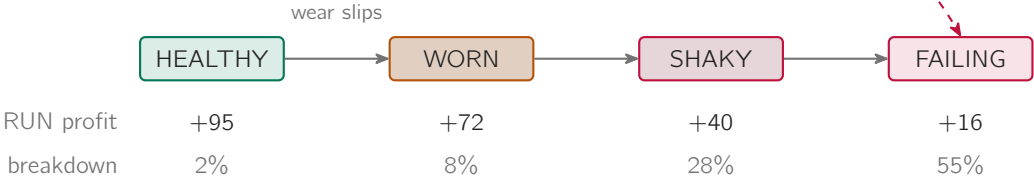
The fleet sheet



The fleet sheet

breakdown under RUN: one net reward, profit minus 280

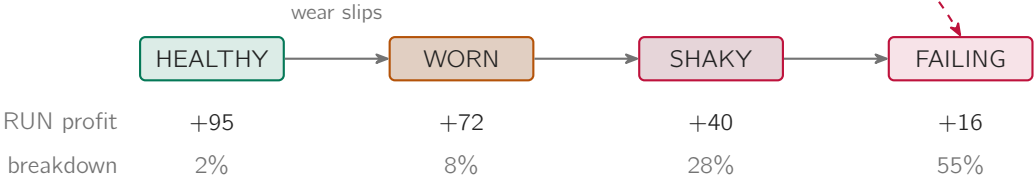
from HEALTHY: $95 - 280 = -185$



The fleet sheet

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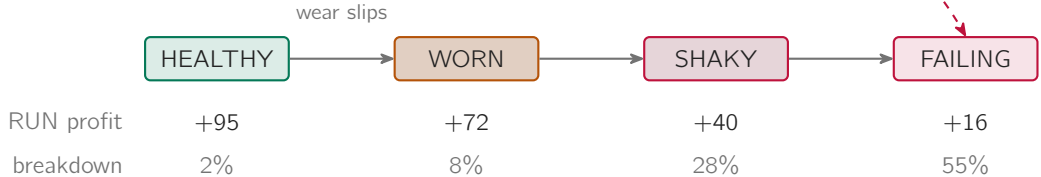


SERVICE: -50, a week in the shop (WORN back to HEALTHY 95% of the time)

The fleet sheet

breakdown under RUN: one net reward, profit minus 280

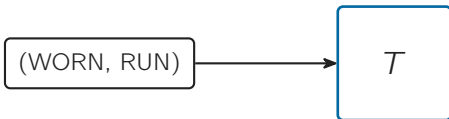
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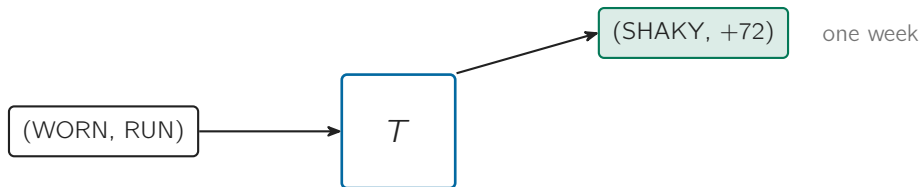
SERVICE: -50, a week in the shop (WORN back to HEALTHY 95% of the time)

REPLACE: -130, a brand-new HEALTHY van next week

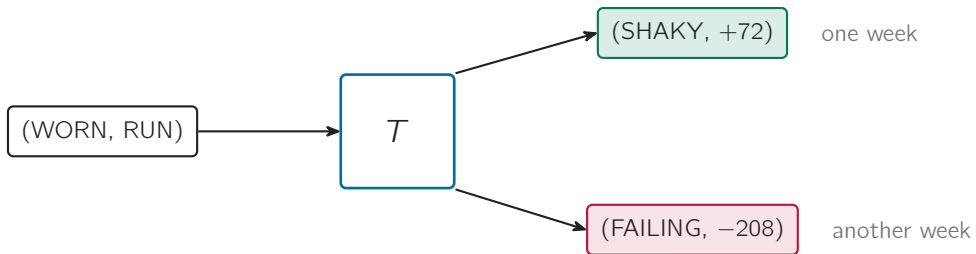
Same pair in, different week out



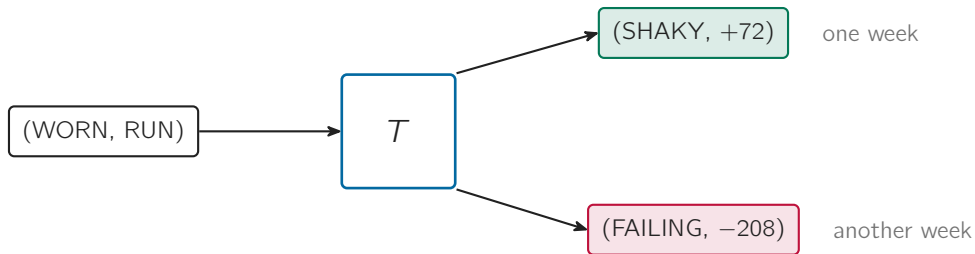
Same pair in, different week out



Same pair in, different week out



Same pair in, different week out



same input, different output: the dynamics T is random

Policies and trajectories

Playbook in, tape out

a policy (the playbook): $\pi : S \rightarrow A$, one call per state

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HEALTHY $\rightarrow$ RUN

WORN $\rightarrow$ SERVICE

SHAKY $\rightarrow$ REPLACE

FAILING $\rightarrow$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

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wk 1
HEALTHY, RUN
+95

Playbook in, tape out

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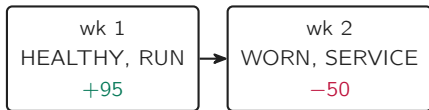
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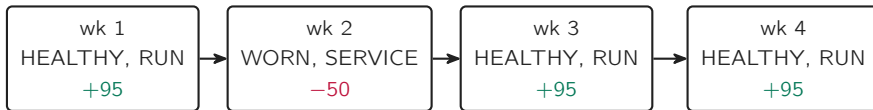


Playbook in, tape out

a policy (the playbook): $\pi : S \rightarrow A$, one call per state

HEALTHY $\rightarrow$ RUN WORN $\rightarrow$ SERVICE SHAKY $\rightarrow$ REPLACE FAILING $\rightarrow$ REPLACE

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each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$



the trajectory: $s_1, a_1, r_1, s_2, a_2, r_2, \dots$ one week per step, forever

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a policy (the playbook): $\pi : S \rightarrow A$, one call per state



each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$



the trajectory: $s_1, a_1, r_1, s_2, a_2, r_2, \dots$ one week per step, forever

the tape is random; another run, same start: (HEALTHY, RUN, -185) $\rightarrow$ FAILING

The goal: accumulating reward

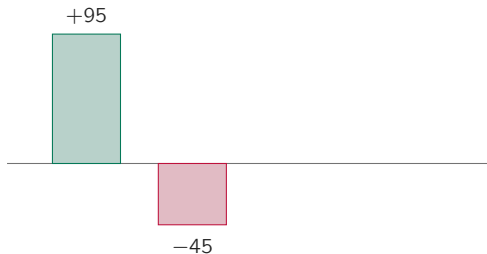
The return: one number per tape

$$G_1 = r_1$$



The return: one number per tape

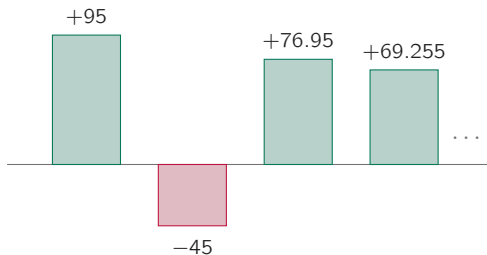
$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma r_2}_{\text{at 90\%}}$$



each week shrunk
by $\gamma = 0.9$

The return: one number per tape

$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma r_2}_{\text{at 90\%}} + \underbrace{\gamma^2 r_3}_{\text{at 81\%}} + \dots$$

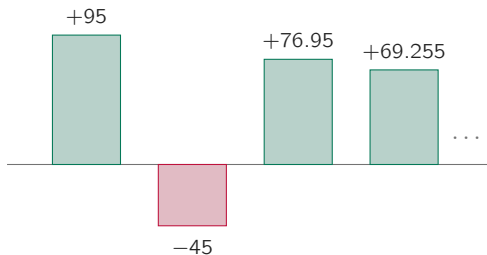


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$$G_1 = 95 + 0.9(-50) + 0.81 \cdot 95 + 0.729 \cdot 95 + \dots = 95 - 45 + 76.95 + 69.255 + \dots$$

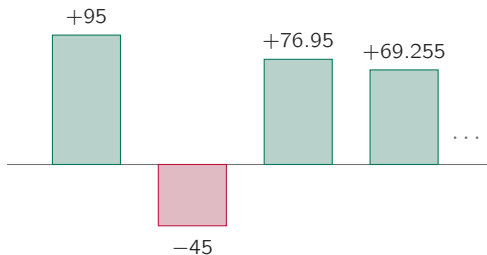


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each week shrunk
by $\gamma = 0.9$

the shrinking factors keep
the never-ending sum finite

The recurrence

$$G_i = r_i + \gamma r_{i+1} + \gamma^2 r_{i+2} + \dots$$

r_i

r_{i+1}

r_{i+2}

$\dots$

The recurrence

$$G_i = r_i + \gamma (r_{i+1} + \gamma r_{i+2} + \dots)$$



G_{i+1} , discounted once by γ

The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



G_{i+1} , discounted once by γ

The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



G_{i+1} , discounted once by γ

this week's reward + $\gamma \times$ all the rest

Maximize the average

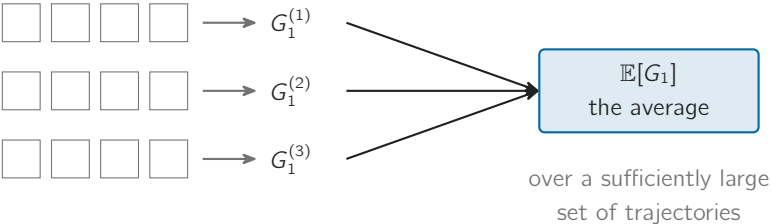
$$\square \square \square \square \longrightarrow G_1^{(1)}$$

$$\square \square \square \square \longrightarrow G_1^{(2)}$$

$$\square \square \square \square \longrightarrow G_1^{(3)}$$

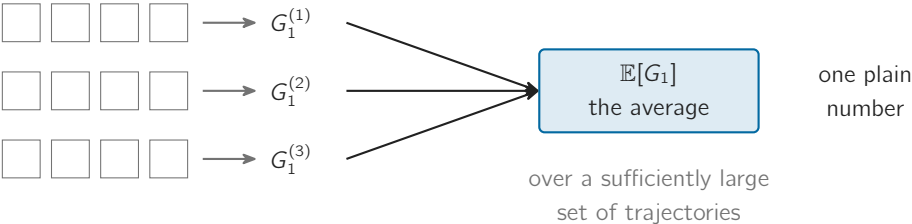
same playbook, same start: a different G_1 every run

Maximize the average



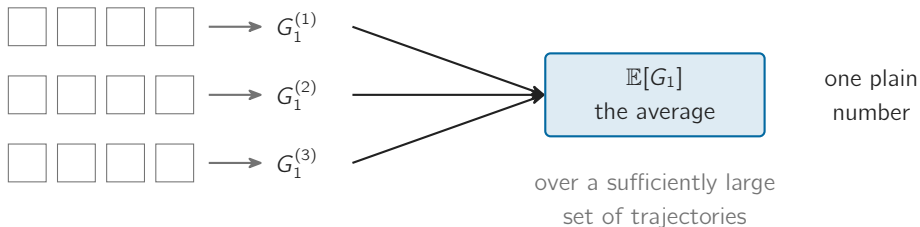
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Maximize the average



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Maximize the average



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the goal: find the policy π that maximizes $\mathbb{E}[G_1]$

The Q -function and the Bellman equation

The scorecard Q^*

$$Q^*(s, a) = \mathbb{E}[G_i \mid s_i = s, a_i = a]$$

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	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9

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	RUN	SERVICE	REPLACE
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FAILING	-3.1	86.9	149.9

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π^* : read each row, keep the call with the largest value

	RUN	SERVICE	REPLACE	
HEALTHY	311.0	229.9	149.9	row by row: the kept call
WORN	203.4	226.1	149.9	
SHAKY	96.5	126.4	149.9	
FAILING	-3.1	86.9	149.9	

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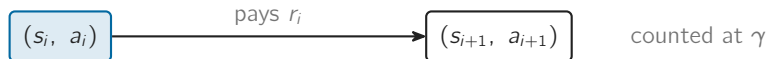
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the optimal playbook replaces a van that still starts every morning

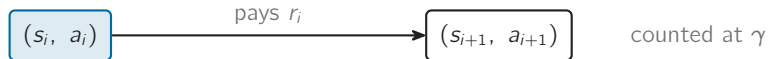
The Bellman equation

$$G_i = r_i + \gamma G_{i+1} \quad (\text{every single week})$$



The Bellman equation

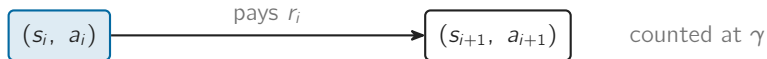
$$Q^*(s_i, a_i) = \mathbb{E}[r_i + \gamma G_{i+1}] \quad (\text{average both sides})$$



The Bellman equation

$$Q^*(s_i, a_i) = \mathbb{E} \left[\underbrace{r_i}_{\text{this week's cash}} + \gamma \underbrace{Q^*(s_{i+1}, a_{i+1})}_{\text{next week's value}} \right]$$

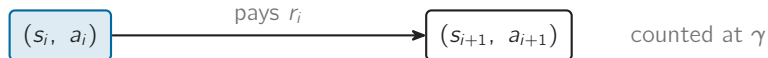
replacing G_{i+1} by its average $Q^*(s_{i+1}, a_{i+1})$: averaging in two rounds



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replacing G_{i+1} by its average $Q^*(s_{i+1}, a_{i+1})$: averaging in two rounds



one equation per cell: twelve equations pin down the table

SARSA: learning from the logbook

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays +72

Overwriter
card = last week 72

Filing cabinet
true average of every week 72

Nudger
move 15% toward the week 72

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72
------------------	-----	-----

Overwriter card = last week	72	72
--------------------------------	----	----

Filing cabinet true average of every week	72	72
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Nudger move 15% toward the week	72	72
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Three bookkeepers, one bad week

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(WORN, RUN) pays	+72	+72	+72
Overwriter card = last week	72	72	72
Filing cabinet true average of every week	72	72	72
Nudger move 15% toward the week	72	72	72

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	-208
Overwriter card = last week	72	72	72	-208
Filing cabinet true average of every week	72	72	72	2
Nudger move 15% toward the week	72	72	72	30

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	-208	+72
Overwriter card = last week	72	72	72	-208	72
Filing cabinet true average of every week	72	72	72	2	16
Nudger move 15% toward the week	72	72	72	30	36.3

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	-208	+72	
Overwriter card = last week	72	72	72	-208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	-208	+72	
Overwriter card = last week	72	72	72	-208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha [r_i - q(s_i, a_i)] \quad (\text{the Nudger, written down})$$

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	-208	+72
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Overwriter card = last week	72	72	72	-208	72	hysterical
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Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
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$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{the card}} + \underbrace{\alpha}_{0.15} \underbrace{[r_i - q(s_i, a_i)]}_{\text{the surprise}}$$

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	-208	+72
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nudge, never overwrite: one bad week must not erase a year

The alpha knob

same five weeks, three step sizes

The alpha knob

same five weeks, three step sizes

$$\alpha = 0.9$$

72



-180



46.8

the -208 week lands

then +72 lands

whiplash

The alpha knob

same five weeks, three step sizes

$$\alpha = 0.9$$

72



-180



46.8

whiplash

the -208 week lands

then +72 lands

$$\alpha = 0.05$$

72



58

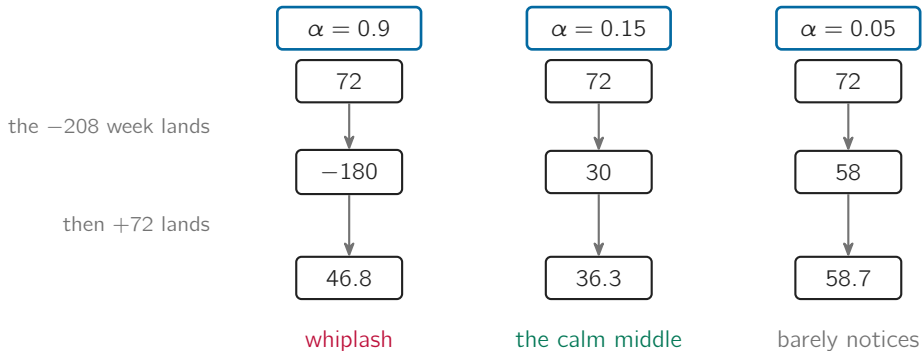


58.7

barely notices

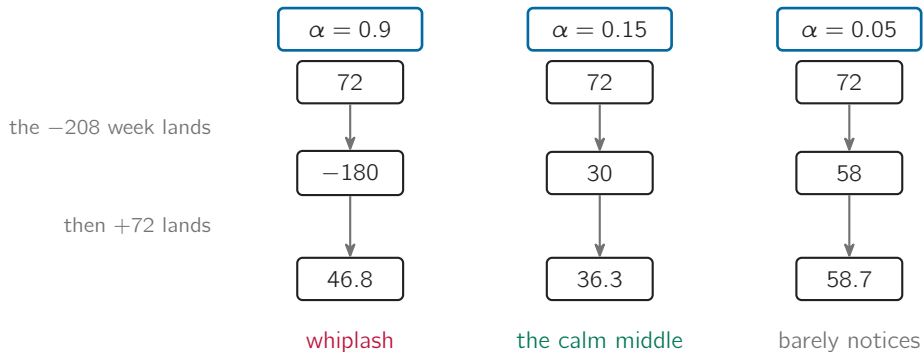
The alpha knob

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The alpha knob

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small steady steps, never overwrite

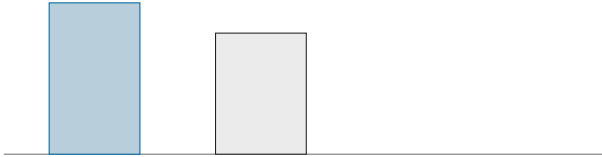
The planner's Monday

Monday forecast: 100



The planner's Monday

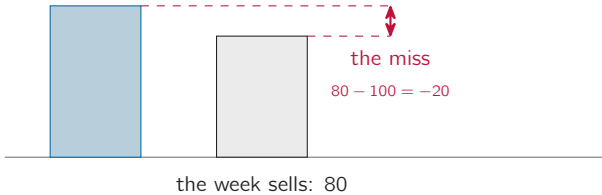
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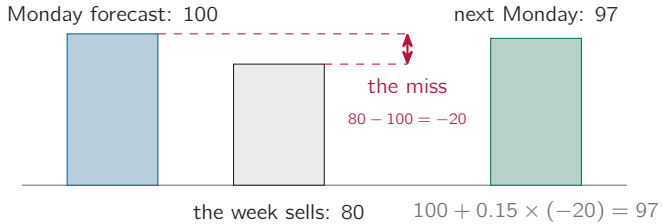
the week sells: 80

The planner's Monday

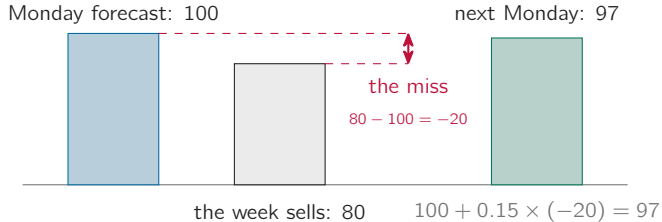
Monday forecast: 100



The planner's Monday

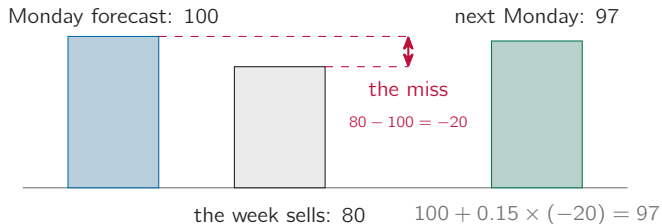


The planner's Monday



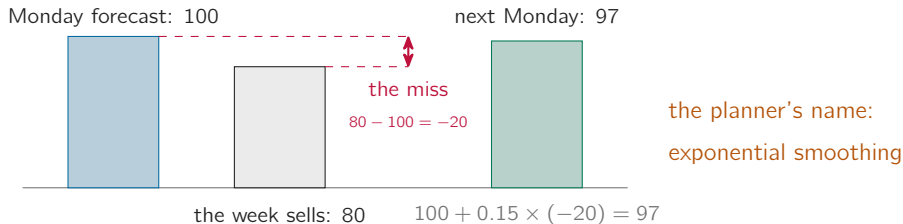
$$\text{new forecast} = \text{old forecast} + 0.15 \times \text{miss}$$

The planner's Monday



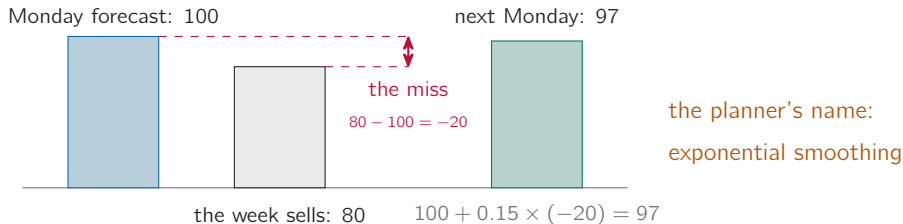
$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

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SARSA forecasts the value of a call the way a planner forecasts sales

The cash-only ledger lies

the cash-only ledger

REPLACE: -130, every single week

worst call on the sheet

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SERVICE: -50

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the scorecard Q^* you saw

REPLACE crowned at SHAKY and FAILING: 149.9

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what does the cash-only ledger miss?

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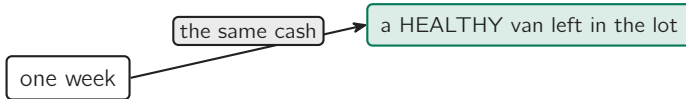
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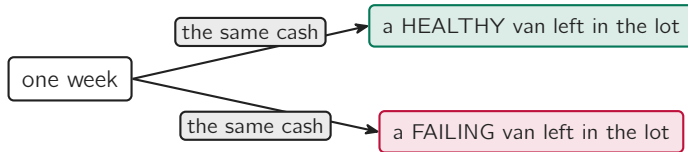
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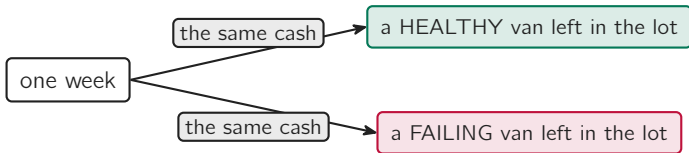
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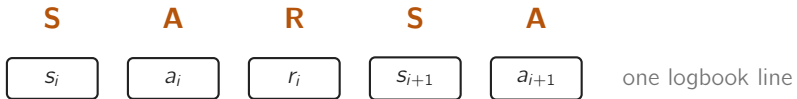
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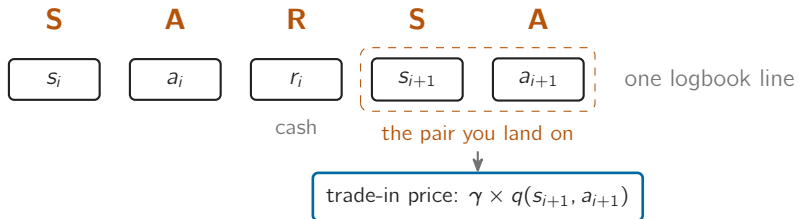


the week leaves you a van: same cash, different futures

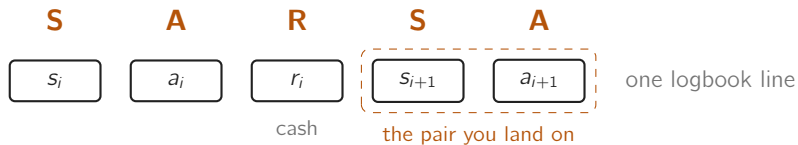
Grade the week like a trade-in



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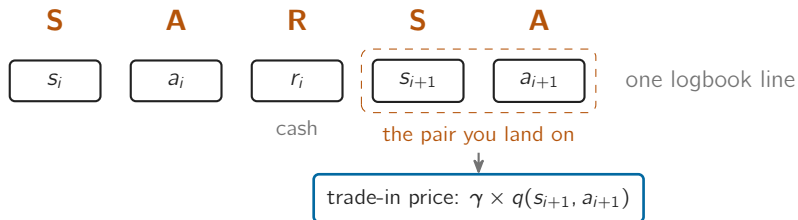
Grade the week like a trade-in



$$\text{the week's worth} = \underbrace{r_i}_{\text{cash}} + \gamma \underbrace{q(s_{i+1}, a_{i+1})}_{\text{trade-in, at your own price}}$$

priced from your own card wall: $q(s_{i+1}, a_{i+1})$ stands in for the unknown Q^* (the NPV move)

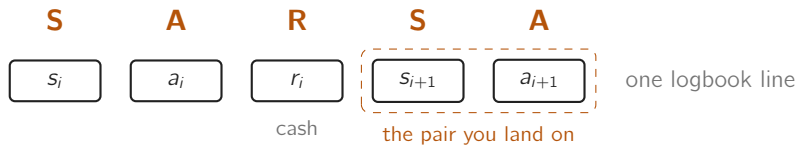
Grade the week like a trade-in



$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha [r_i - q(s_i, a_i)] \quad (\text{cash only})$$

priced from your own card wall: $q(s_{i+1}, a_{i+1})$ stands in for the unknown Q^* (the NPV move)

Grade the week like a trade-in

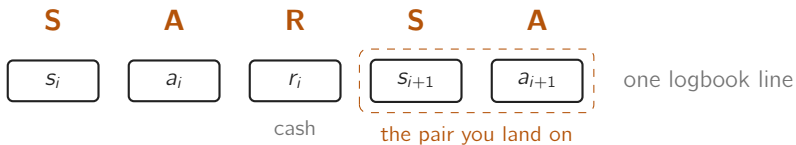


trade-in price: $\gamma \times q(s_{i+1}, a_{i+1})$

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \left[\underbrace{r_i + \gamma q(s_{i+1}, a_{i+1})}_{\text{the actual, grown up}} - q(s_i, a_i) \right]$$

priced from your own card wall: $q(s_{i+1}, a_{i+1})$ stands in for the unknown Q^* (the NPV move)

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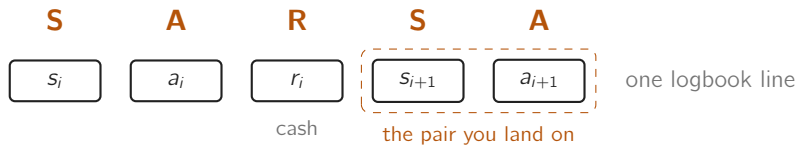


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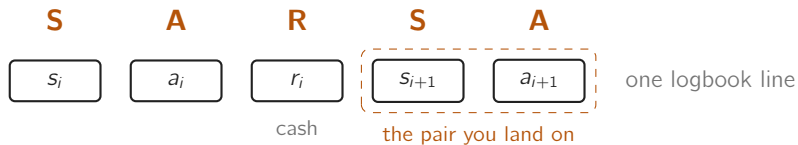
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$\alpha = 0.15$, constant · $\varepsilon = 0.15$: a random call in 15% of weeks · a_{i+1} : actually executed, never a max

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Bellman, with the expectation replaced by an arriving average

One week, one nudge

twelve guesses, all start at 0

	RUN	SERVICE	REPLACE
HEALTHY	0	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

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week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

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$$\text{surprise} = 95 - 0 = 95$$

nudge: $+0.15 \times 95$

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nudge: $+0.15 \times 95$



		RUN	SERVICE	REPLACE
twelve guesses, all start at 0	HEALTHY	14.25	0	0
one cell moves, eleven stay put	WORN	0	0	0
	SHAKY	0	0	0
	FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

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twelve guesses, all start at 0	HEALTHY	14.25	0	0
one cell moves, eleven stay put	WORN	0	0	0
	SHAKY	0	0	0
a few hundred weeks later: the three bands emerge	FAILING	0	0	0

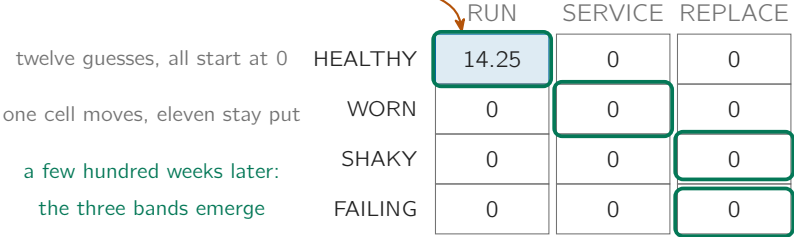
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one cell moves, eleven stay put	WORN	0	0	0
a few hundred weeks later:	SHAKY	0	0	0
the three bands emerge	FAILING	0	0	0

the cells settle below Q^* and keep jittering, but each row keeps the right call

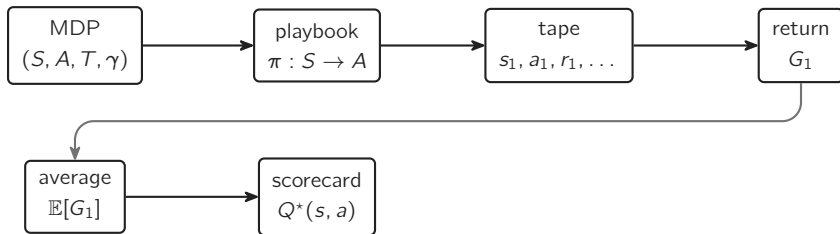
The whole story



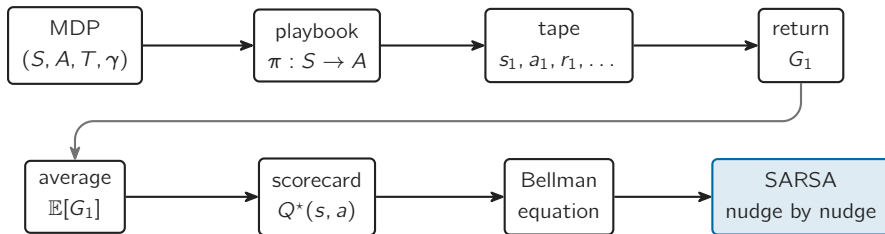
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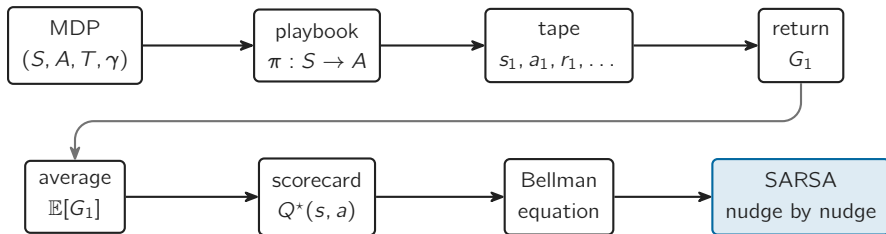
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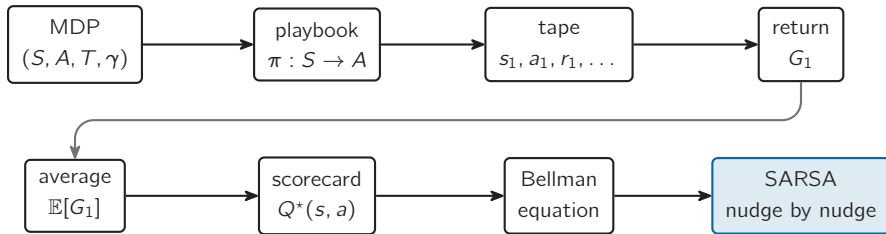


The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

play it: https://sml-fs26.github.io/classic_rl/repair-or-replace/

Appendix: why the nudge works

The Robbins-Monro iteration

find the number x^* that equals the average of noisy samples, one at a time

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SARSA is Robbins-Monro on the Bellman equation, cell by cell

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A_n : the average of the first n observations $x_1, \dots, x_n$

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SARSA is a running average of the weekly evidence